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Print Ease: A Smart Campus Printing System Developed Using C++ Object-Oriented Programming and Priority Queue Management

DSA – 0108

Object oriented programming using c++

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Abstract



- Print Ease automates the campus printing process by replacing manual printing requests with a simple digital platform for students, faculty, and staff.
- The system uses C++ Object-Oriented Programming and Priority Queue to organize, manage, and process print jobs efficiently based on their assigned priority.
- Users can securely authenticate, submit documents, estimate printing costs, select printing options, and track the current status of their print requests.
- Administrators can monitor print jobs, manage users, control the printing queue, update job statuses, and generate reports for centralized campus printing management.
- The key idea is Digital Printing → Priority Scheduling → Faster Processing → Better Management, resulting in an organized, efficient, and user-friendly campus printing system.



Introduction

- Campus printing is an essential service used regularly by students, faculty, and staff for academic and administrative purposes.
- Traditional manual printing processes often result in long queues, waiting times, confusion, and delays during busy campus hours.
- Handling multiple print requests manually can cause inefficient job processing, incorrect order management, and difficulty in tracking requests.
- Print Ease provides a digital platform for print request submission, job scheduling, cost estimation, status tracking, and centralized print management.
- The Priority Queue ensures that important and urgent print jobs are processed efficiently, improving speed, organization, and overall printing service management.



Literature Survey



S.No	Authors & Year	Journal / Conference	Technique / Contribution	Limitations
1	Mustafa et al. (2025)	International Journal of Computer Applications	Cloud-based automated print queue management	LIFO is not suitable for priority/urgent jobs.
2	Dhabliya et al. (2025)	ShodhKosh Journal	Cloud-based print management with secure printing	Depends on network and has security challenges.
3	Li et al. (2024)	IEEE Access	Print-job sequence optimization and scheduling	Mainly designed for industrial printing systems.
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Objectives



- Develop a **C++ OOP-based digital campus printing system** to simplify and automate the overall printing process.
- Implement a **Priority Queue scheduling mechanism** to arrange and process print jobs according to their assigned priority levels.
- Automate the complete **print-job management process**, including request submission, job scheduling, processing, and status tracking.
- **Reduce user waiting time** by efficiently organizing print requests and processing high-priority jobs without unnecessary delays.
- Improve **centralized administrative control** by providing administrators with effective monitoring, management, and reporting features.



Existing System



- Print requests are submitted through **manual methods**, making the overall printing process time-consuming and less convenient.
- **Long waiting times** occur during peak hours because many students and faculty members depend on the same printing service.
- There is **no priority-based scheduling**, so urgent print jobs may not be processed before regular requests.
- The existing process provides **limited print-order status tracking**, making it difficult for users to know whether their documents are pending, processing, or completed.
- Managing **multiple print requests** manually becomes difficult and can result in **errors, confusion, and processing delays**.



Proposed System



- Provides **digital submission of print requests**, allowing students and faculty to submit printing requirements conveniently through the system.
- Supports **document upload and print options**, enabling users to select required settings such as number of copies and printing preferences.
- Performs **automatic cost calculation** based on selected print options, helping users know the estimated printing cost before processing.
- Uses a **Priority Queue** to schedule and process print jobs efficiently according to their assigned priority levels.
- Provides **real-time print-status tracking**, allowing users to monitor whether their print request is pending, processing, or completed.
- Includes **admin monitoring and report generation** to help administrators manage print jobs, users, and overall printing activities effectively.



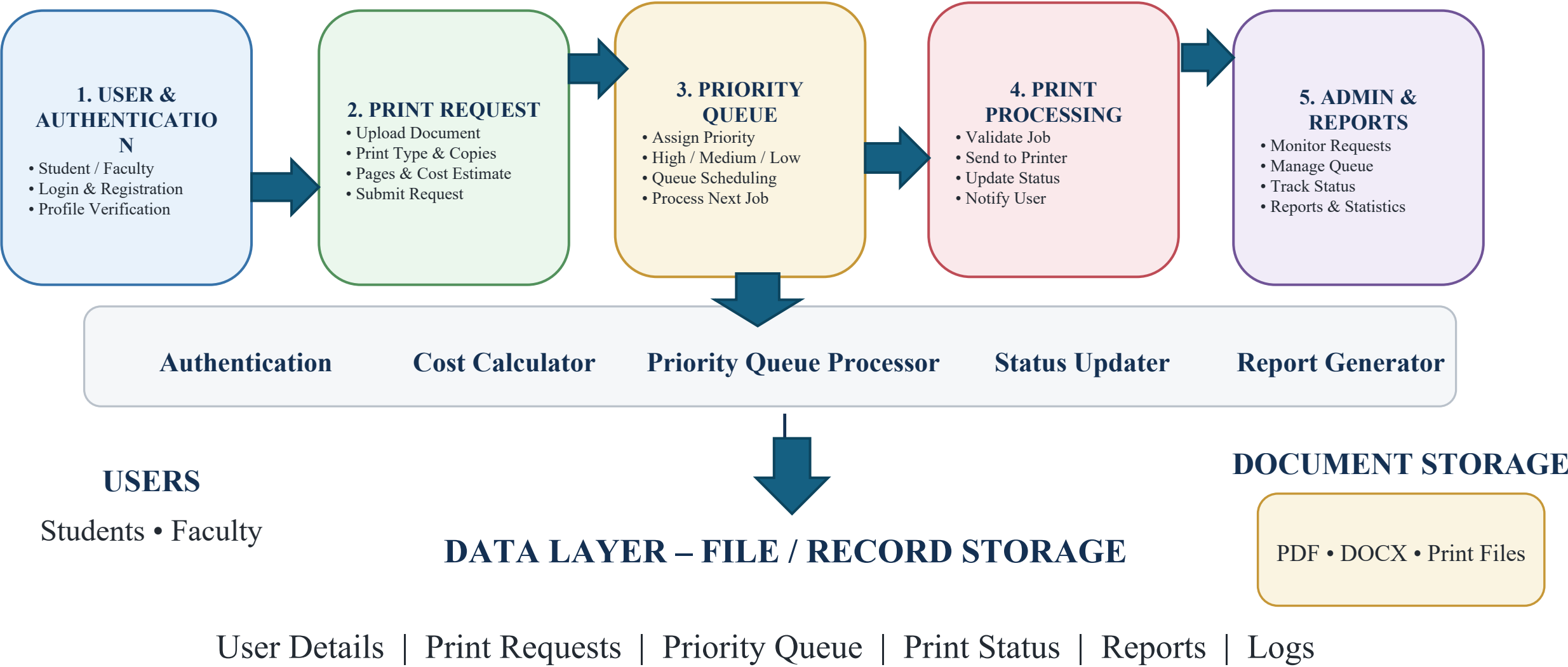
Problem Statement



- **Manual print request handling** creates delays and long waiting times, especially when the number of printing requests increases during peak campus hours.
- The absence of a **priority-based scheduling system** makes it difficult
 - to process urgent and faculty print jobs before regular printing requests.
- **Multiple print requests** are difficult to organize manually, which can result in confusion, incorrect processing order, and unnecessary delays.
- Limited **print-job tracking** prevents users from easily monitoring the current status of their submitted printing requests.
- Lack of proper **administrative monitoring and centralized control** reduces overall operational efficiency and makes print-service management more difficult.



SYSTEM ARCHITECTURE





Technology Stack



- **C++17** — Used for the core backend, business logic, object-oriented programming, and print-job management.
- **HTML5** — Provides the basic frontend for user login, print requests, dashboards, and status tracking.
- **CSS3** — Used to design the user interface, including layouts, styling, spacing, and responsive elements.
- **JavaScript (ES6)** — Handles frontend interactions, form validation, dynamic updates, and communication with the backend.
- **WinSock2** — Used to implement a custom HTTP server in C++ for communication between the frontend and backend.



Data Structures & Algorithms



- **Priority Queue (Binary Max-Heap)** — Serves as the core scheduling mechanism, ensuring high-priority print jobs are selected and processed first.
- **FIFO Queue** — Manages normal-priority print requests in the order they are received, providing fair processing for regular jobs.
- **Stack (LIFO)** — Maintains recently completed print orders and history, allowing the latest completed request to be accessed quickly.
- **Priority Scheduling** — Assigns priority levels to print jobs so that urgent and important requests are processed before lower-priority requests.
- **Adaptive Priority Scheduling** — Dynamically increases the priority of waiting print jobs based on factors such as **waiting time, urgency, and user category**, helping prevent starvation and improve overall fairness.



Module 1 — User & Authentication

- **Purpose**
- Manages user registration, login, authentication, and profile information.
- **Functions**
- Student/User registration
- Secure login
- User authentication
- Password validation
- Profile management
- View personal details
- Logout
- Store user information



Create Account

Get started with PrintEase today

Student

Shopkeeper

Username

e.g. SaranRaj

Email Address

you@example.com

Hostel

Please fill in this field.

ROOM NUMBER

Select hostel ▼

e.g. 204

Password

Create a strong password

Create Account



Module 2 – Print Request Management



- Upload Print Document (PDF, DOCX, etc.)
- Select Document Type
- Choose Print Mode (Black & White / Color)
- Enter Number of Copies
- Enter Number of Pages
- Automatic Cost Calculation
- Submit Print Request
- Save Print Request History



MENU

Dashboard

New Order

My Orders

Track Order

ACCOUNT

Profile

Logout

Upload Documents

Drag & drop PDF files here

Or click to browse • PDF only • Max 25MB each • Multiple files supported

Document Classification & Priority

Document Type (Determines Base Priority in Heap)

Hall Ticket (100 pts)

Exam Record (75 pts)

Assignment (50 pts)

Others (25 pts)

Urgency Level

Normal Scheduling (+0 pts)

Urgent (Exam in ≤2 hrs) (+50 pts)

Print Options

Print Type

Black & White (₹2/pg)

Color (₹10/pg)

Print Side

Single Side

Double Side (₹1.5/pg)

Number of Copies

1



Module 3 – Priority Queue Scheduling



- Receives print requests from students.
- Assigns priority levels to each print job.
- Stores requests using a **Priority Queue**.
- Processes higher-priority jobs before lower-priority jobs.
- Displays the current print queue.
- Updates print job status (Pending → Printing → Completed).
- Automatically rearranges the queue when new requests arrive.
- Reduces waiting time through efficient scheduling.
- Ensures fair and organized print job execution.

⚡ Document Classification & Priority

Document Type (Determines Base Priority in Heap)

📄 Hall (100
Ticket pts)

📄 Exam (75
Record pts)

📄 Assignment (50
pts)

📄 Others (25
pts)

Urgency Level

Normal Scheduling (+0 pts)

🚨 Urgent (Exam in ≤2 hrs) (+50 pts)



Module 4 – Admin & Report Management



- Secure Admin Login
- View All Print Requests
- Monitor Pending Jobs
- Start Print Process
- Update Print Status
- Manage Student Records
- View Completed Jobs
- View Printing Statistics
- Search Print Requests
- Delete Completed Records (Optional)





CONCLUSION



- Print Ease provides a digital solution that simplifies and automates the campus printing process for students and faculty.
- The integration of C++ Object-Oriented Programming and Priority Queue enables efficient organization and processing of multiple print requests.
- Adaptive Priority Scheduling improves the system by giving higher importance to urgent jobs while preventing long-waiting requests from being ignored.
- Features such as authentication, document submission, cost calculation, status tracking, admin monitoring, and report generation provide complete print-job management.
- Overall, Print Ease reduces waiting time, improves printing efficiency, provides better administrative control, and creates a faster and more organized campus printing experience.